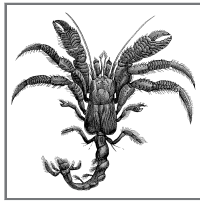


6
Graph Algorithms

Graphs are fundamental structures that represent complex structured information. The images in Figure 6-1 are all sample graphs.

In this chapter, we investigate common ways to represent graphs and associated algorithms that frequently occur. Inherently, a graph contains a set of elements, known as *vertices*, and relationships between pairs of these elements, known as *edges*. We use these terms consistently in this chapter; other descriptions might use the terms "node" and "link" to represent the same information. We consider only simple graphs that avoid (a) self-edges from a vertex to itself, and (b) multiple edges between the same pair of vertices.

Given the structure defined by the edges in a graph, many problems can be stated in terms of *paths* from a source vertex to a destination vertex in the graph constructed using the existing edges in the graph. Sometimes an edge has an associated numeric value known as its *weight*; sometimes an edge is *directed* with a specific orientation (like a one-way street). In the **Single-Source Shortest Path** algorithm, one is given a specific vertex, *s*, and asked to compute the shortest path (by sum of edge weights) to all other vertices in the graph. The **All-Pairs Shortest Path** problem requires that the shortest path be computed for all pairs (*u*, *v*) of vertices in the graph.

Some problems seek a deeper understanding of the underlying graph structure. For example, the minimum spanning tree (MST) of an undirected, weighted graph is a subset of that graph's edges such that (a) the original set of vertices is still connected in the MST, and (b) the sum total of the weights of the edges in the MST is minimum. We show how to efficiently solve this problem using **Prim's Algorithm**.

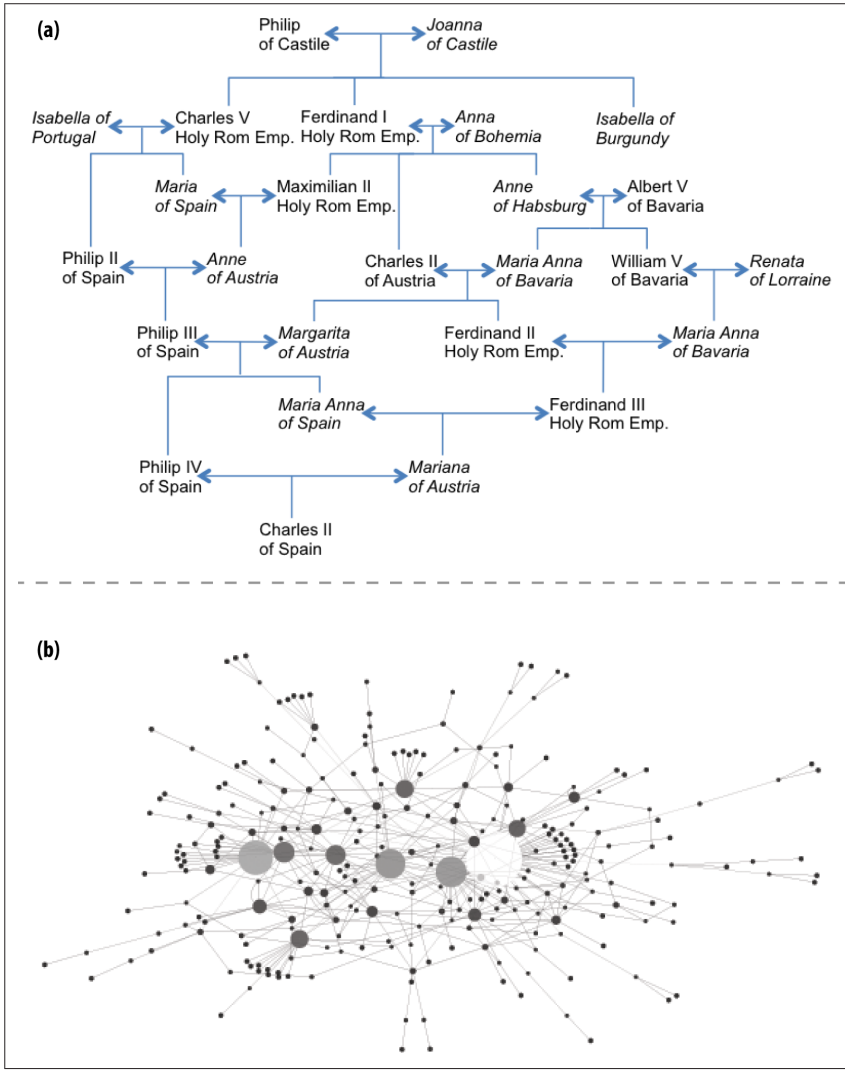


Figure 6-1. (a) The genealogy of Charles II of Spain (1661–1700); (b) a molecular network related to liver cancer

Graphs

A graph G = (*V*, *E*) is defined by a set of vertices, *V*, and a set of edges, *E*, over pairs of these vertices. There are three common types of graphs:

Undirected, unweighted graphs

These model relationships between vertices (*u*, *v*) without regard for the direction of the relationship. These graphs are useful for capturing symmetric infor-

mation. For example, in a graph modeling a social network, if *Alice* is a friend of *Bob*, then *Bob* is a friend of *Alice*.

Directed graphs
: These model relationships between vertices (u, v) that are distinct from the relationship between (v, u), which may or may not exist. For example, a program to provide driving directions must store information on one-way streets to avoid giving illegal directions.

Weighted graphs
: These model relationships where there is a numeric value known as a *weight* associated with the relationship between vertices (u, v). Sometimes these values can store arbitrary non-numeric information. For example, the edge between towns A and B could store the mileage between the towns, the estimated traveling time in minutes, or the name of the street or highway connecting the towns.

The most highly structured of the graphs—a directed, weighted graph—defines a nonempty set of vertices $\{v_0, v_1, \ldots, v_{n-1}\}$, a set of directed edges between pairs of distinct vertices (such that every pair has at most one edge between them in each direction), and a positive weight associated with each edge. In many applications, the weight is considered to be a distance or cost. For some applications, we may want to relax the restriction that the weight must be positive (e.g., a negative weight could reflect a loss, not a profit), but we will be careful to declare when this happens.

Consider the directed, weighted graph in Figure 6-2, which is composed of six vertices and five edges. We could store the graph using adjacency lists, as shown in Figure 6-3, where each vertex v_i maintains a linked list of nodes, each of which stores the weight of the edge leading to an adjacent vertex of v_i. Thus, the base structure is a one-dimensional array of vertices in the graph.

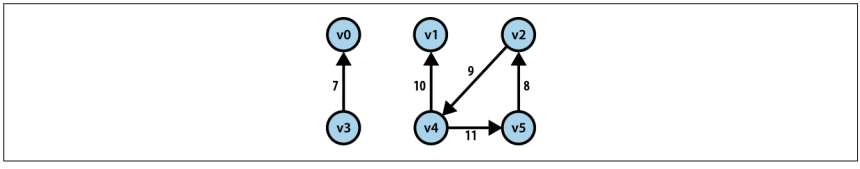


Figure 6-2. Sample directed, weighted graph

Figure 6-4 shows how to store the directed, weighted graph as an n-by-n adjacency matrix A of integers, indexed in both dimensions by the vertices. The entry $A[i][j]$ stores the weight of the edge from v_i to v_j; when there is no edge from v_i to v_j, $A[i][j]$ is set to some special value, such as 0, –1 or even $-\infty$. We can use adjacency lists and matrices to store unweighted graphs as well (perhaps using the value 1 to represent an edge). With an adjacency matrix, checking whether an edge (v_i, v_j) exists takes constant time, but with an adjacency list, it depends on the number of edges in the list for v_i. In contrast, with an adjacency matrix, you need more space and you lose

the ability to identify all incident edges to a vertex *in time proportional to the number of those edges*; instead, you must check all possible edges, which becomes significantly more expensive when the number of vertices becomes large. You should use an adjacency matrix representation when working with *dense graphs* in which nearly every possible edge exists.

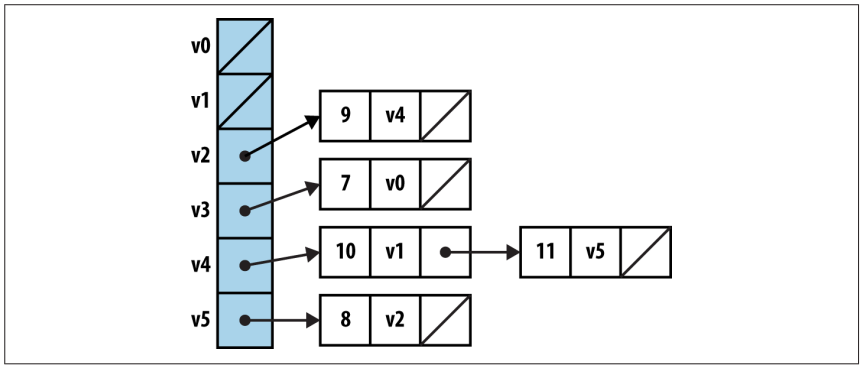


Figure 6-3. Adjacency list representation of directed, weighted graph

We use the notation $<v_0, v_1, \ldots, v_{k-1}>$ to describe a path of k vertices in a graph that traverses $k - 1$ edges (v_i, v_{i+1}) for $0 \leq i < k - 1$; paths in a directed graph honor the direction of the edge. In Figure 6-2, the path $<v_4, v_5, v_2, v_4, v_1>$ is valid. In this graph there is a *cycle*, which is a path of vertices that includes the same vertex multiple times. A cycle is typically represented in its most minimal form. If a path exists between any two pairs of vertices in a graph, then that graph is *connected*.

	v0	v1	v2	v3	v4	v5
v0	0	0	0	0	0	0
v1	0	0	0	0	0	0
v2	0	0	0	0	9	0
v3	7	0	0	0	0	0
v4	0	10	0	0	0	11
v5	0	0	8	0	0	0

Figure 6-4. Adjacency matrix representation of directed, weighted graph

When using an adjacency list to store an undirected graph, the same edge (u, v) appears twice: once in the linked list of neighbor vertices for u and once for v. Thus, undirected graphs may require up to twice as much storage in an adjacency list as a directed graph with the same number of vertices and edges. Doing so lets you locate the neighbors for a vertex u in time proportional to the number of actual neighbors.

When using an adjacency matrix to store an undirected graph, entry $A[i][j] = A[j][i]$.

Data Structure Design

We implement a C++ `Graph` class to store a directed (or undirected) graph using an adjacency list representation implemented with core classes from the C++ Standard Template Library (STL). Specifically, it stores the information as an array of `list` objects, with one `list` for each vertex. For each vertex u there is a `list` of `IntegerPair` objects representing the edge (u, v) of weight w.

The operations on graphs are subdivided into several categories:

Create
: A graph can be initially constructed from a set of n vertices, and it may be directed or undirected. When a graph is undirected, adding edge (u, v) also adds edge (v, u).

Inspect
: We can determine whether a graph is directed, find all incident edges to a given vertex, determine whether a specific edge exists, and determine the weight associated with an edge. We can also construct an iterator that returns the neighboring edges (and their weights) for any vertex in a graph.

Update
: We can add edges to (or remove edges from) a graph. It is also possible to add a vertex to (or remove a vertex from) a graph, but algorithms in this chapter do not need to add or remove vertices.

We begin by discussing ways to explore a graph. Two common search strategies are **Depth-First Search** and **Breadth-First Search**.

Depth-First Search

Consider the maze shown on the left in Figure 6-5. After some practice, a child can rapidly find the path that stretches from the start box labeled s to the target box labeled t. One way to solve this problem is to make as much forward progress as possible and randomly select a direction whenever a choice is possible, marking where you have come from. If you ever reach a dead end or you revisit a location you have already seen, then backtrack until an untraveled branch is found and set off in that direction. The numbers on the right side of Figure 6-5 reflect the branching points of one such solution; in fact, every square in the maze is visited in this solution.

We can represent the maze in Figure 6-5 by creating a graph consisting of vertices and edges. A vertex is created for each branching point in the maze (labeled by numbers on the right in Figure 6-5) as well as "dead ends." An edge exists only if there is a direct path in the maze between the two vertices where no choice in direc-

tion can be made. The undirected graph representation of the maze from Figure 6-5 is shown in Figure 6-6; each vertex has a unique identifier.

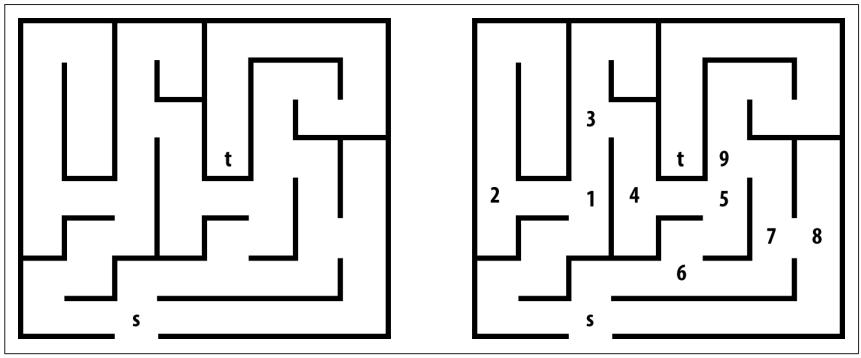


Figure 6-5. A small maze to get from s to t

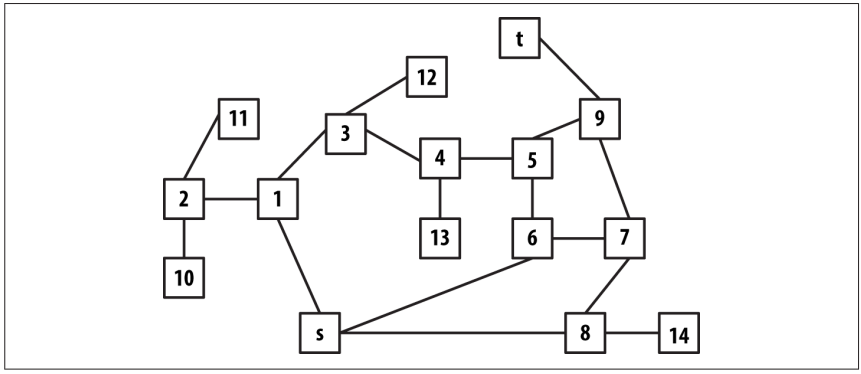


Figure 6-6. Graph representation of maze from Figure 6-5

To solve the maze, we need only find a path in the graph $G = (V, E)$ of Figure 6-5 from the start vertex, *s*, to the target vertex, *t*. In this example, all edges are undirected, but we could easily consider directed edges if the maze imposed such restrictions.

The heart of **Depth-First Search** is a recursive `dfsVisit(u)` operation that visits a vertex *u* that has not yet been visited. `dfsVisit(u)` records its progress by coloring vertices one of three colors:

White
: Vertex has not yet been visited.

Gray
: Vertex has been visited, but it may have an adjacent vertex that has not yet been visited.

Black
: Vertex has been visited and so have all of its adjacent vertices.

Depth-First Search Summary

Best, Average, Worst: O(*V*+*E*)

```
depthFirstSearch (G,s)
  foreach v in V do
    pred[v] = -1
    color[v] = White ❶
  dfsVisit(s)
end

dfsVisit(u)
  color[u] = Gray
  foreach neighbor v of u do
    if color[v] = White then ❷
      pred[v] = u
      dfsVisit(v)
  color[u] = Black ❸
end
```

❶ Initially all vertices are marked as not visited.

❷ Find unvisited neighbor and head in that direction.

❸ Once all neighbors are visited, this vertex is done.

Initially, each vertex is colored white to represent that it has not yet been visited, and **Depth-First Search** invokes `dfsVisit` on the source vertex, *s*. `dfsVisit(u)` colors *u* gray before recursively invoking `dfsVisit` on all adjacent vertices of *u* that have not yet been visited (i.e., they are colored white). Once these recursive calls have completed, *u* can be colored black, and the function returns. When the recursive `dfsVisit` function returns, **Depth-First Search** backtracks to an earlier vertex in the search (indeed, to a vertex that is colored gray), which may have an unvisited adjacent vertex that must be explored. Figure 6-7 contains an example showing the partial progress on a small graph.

For both directed and undirected graphs, **Depth-First Search** investigates the graph from *s* until all vertices reachable from *s* are visited. During its execution, **Depth-First Search** traverses the edges of the graph, computing information that reveals the inherent, complex structure of the graph. For each vertex, **Depth-First Search** records `pred[v]`, the predecessor vertex to *v* that can be used to recover a path from the source vertex *s* to the vertex *v*.

This computed information is useful to a variety of algorithms built on **Depth-First Search**, including topological sort and identifying strongly connected components.

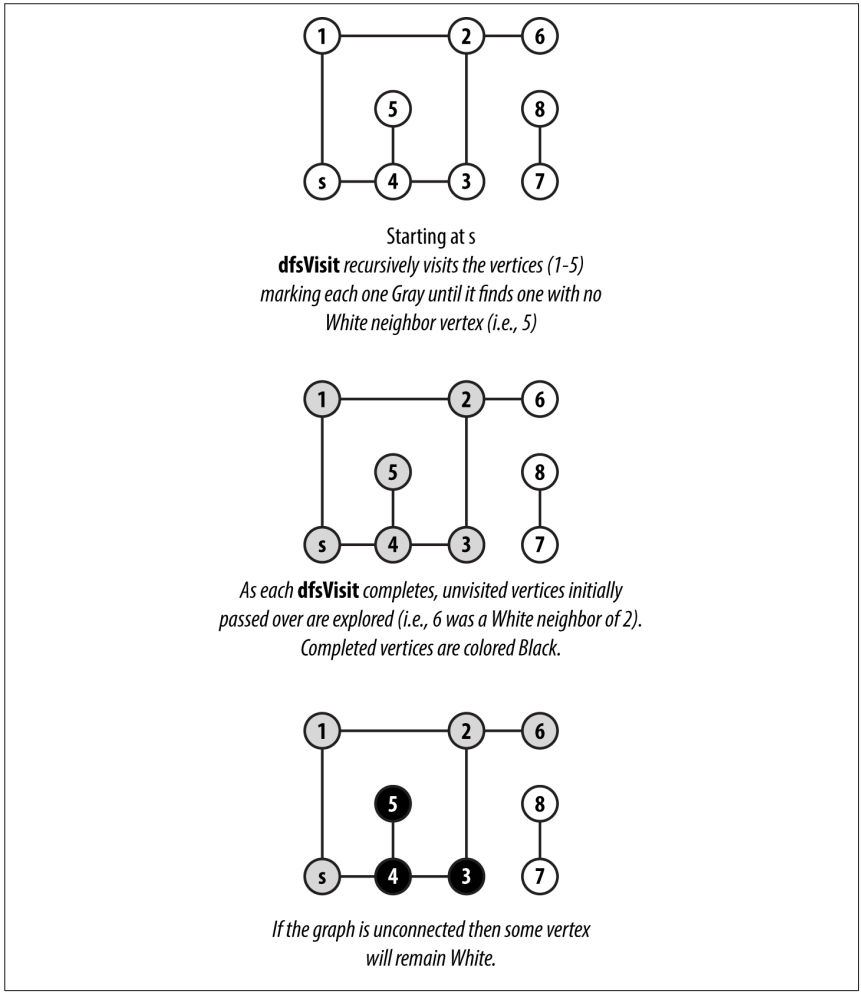
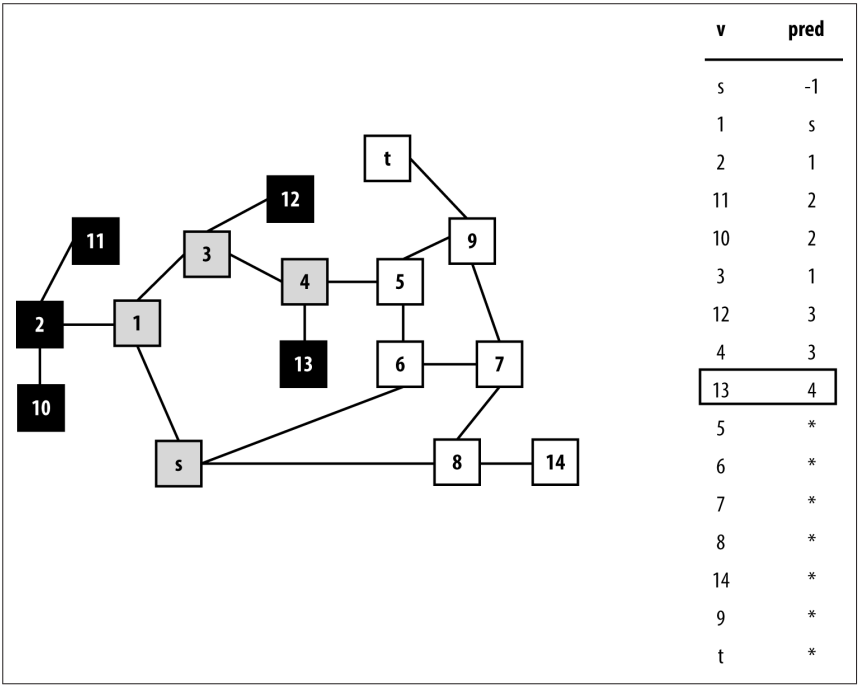


Figure 6-7. Depth-First Search example

Given the graph in Figure 6-6 and assuming the neighbors of a vertex are listed in increasing numerical order, the information computed during the search is shown in Figure 6-8. The coloring of the vertices of the graph shows the snapshot just after the fifth vertex (in this case, vertex 13) is colored black. Some parts of the graph (i.e., the vertices colored black) have been fully searched and will not be revisited. Note that white vertices have not been visited yet and gray vertices are currently being recursively visited by `dfsVisit`.

Depth-First Search has no global awareness of the graph, and so it will blindly search the vertices <5, 6, 7, 8>, even though these are in the wrong direction from the target, *t*. Once **Depth-First Search** completes, the `pred[]` values can be used to generate a path from the original source vertex, *s*, to each vertex in the graph.



v	pred
s	-1
1	s
2	1
11	2
10	2
3	1
12	3
4	3
13	4
5	*
6	*
7	*
8	*
14	*
9	*
t	*

Figure 6-8. Computed pred for a sample undirected graph; snapshot taken after five vertices are colored black

Note that this path may not be the shortest possible path; when **Depth-First Search** completes, the path from *s* to *t* has seven vertices <*s*, 1, 3, 4, 5, 9, *t*>, while a shorter path of five vertices exists <*s*, 6, 5, 9, *t*>. Here the notion of a "shortest path" refers to the number of decision points between *s* and *t*.

Input/Output

The input is a graph $G = (V, E)$ and a source vertex $s \in V$ representing the start location.

Depth-First Search produces the `pred[v]` array that records the predecessor vertex of *v* based on the depth-first search ordering.

Context

Depth-First Search only needs to store a color (either white, gray, or black) with each vertex as it traverses the graph. Thus, **Depth-First Search** requires only $O(n)$ overhead in storing information while it explores the graph starting from *s*.

Depth-First Search can store its processing information in arrays separately from the graph. **Depth-First Search** only requires that it can iterate over the vertices in a graph that are adjacent to a given vertex. This feature makes it easy to perform

Depth-First Search on complex information, since the `dfsVisit` function accesses the original graph as a read-only structure.

Solution

Example 6-1 contains a sample C++ solution. Note that vertex color information is used only within the `dfsVisit` methods.

Example 6-1. Depth-First Search implementation

```
// visit a vertex, u, in the graph and update information
void dfsVisit (Graph const &graph, int u,                  /* in */
         vector<int> &pred, vector<vertexColor> &color) {  /* out */
  color[u] = Gray;

  // process all neighbors of u.
  for (VertexList::const_iterator ci = graph.begin (u);
       ci != graph.end (u); ++ci) {
    int v = ci->first;

    // Explore unvisited vertices immediately and record pred[].
    // Once recursive call ends, backtrack to adjacent vertices.
    if (color[v] == White) {
      pred[v] = u;
      dfsVisit (graph, v, pred, color);
    }
  }

  color[u] = Black;  // our neighbors are complete; now so are we.
}

/**
 * Perform Depth-First Search starting from vertex s, and compute
 * pred[u], the predecessor vertex to u in resulting depth-first
 * search forest.
 */
void dfsSearch (Graph const &graph, int s,    /* in */
         vector<int> &pred) {                 /* out */
  // initialize pred[] array and mark all vertices White
  // to signify unvisited.
  const int n = graph.numVertices();
  vector<vertexColor> color (n, White);
  pred.assign(n, -1);

  // Search starting at the source vertex.
  dfsVisit (graph, s, pred, color);
}
```

Analysis

The recursive `dfsVisit` function is called once for each vertex in the graph. Within `dfsVisit`, every neighboring vertex must be checked; for directed graphs, edges are traversed once, whereas in undirected graphs they are traversed once and are seen one other time. In any event, the total performance cost is $O(V + E)$.

Variations

If the original graph is unconnected, then there may be no path between *s* and some vertices; these vertices will remain unvisited. Some variations ensure all vertices are processed by conducting additional `dfsVisit` executions on the unvisited vertices in the `dfsSearch` method. If this is done, `pred[]` values record a depth-first forest of depth-first tree search results. To find the roots of the trees in this forest, scan `pred[]` to find vertices *r* whose `pred[r]` value is –1.

Breadth-First Search

Breadth-First Search takes a different approach from **Depth-First Search** when searching a graph. **Breadth-First Search** systematically visits all vertices in the graph $G = (V, E)$ that are k edges away from the source vertex *s* before visiting any vertex that is $k + 1$ edges away. This process repeats until no more vertices are reachable from *s*. **Breadth-First Search** does not visit vertices in *G* that are not reachable from *s*. The algorithm works for undirected as well as directed graphs.

Breadth-First Search is guaranteed to find the shortest path in the graph from vertex *s* to a desired *target* vertex, although it may evaluate a rather large number of nodes as it operates. **Depth-First Search** tries to make as much progress as possible, and may locate a path more quickly, which may not be the shortest path.

Figure 6-9 shows the partial progress of **Breadth-First-Search** on the same small graph from Figure 6-7. First observe that the gray vertices in the graph are exactly the ones contained within the queue. Each time through the loop a vertex is removed from the queue and unvisited neighbors are added.

Breadth-First Search makes its progress without requiring any backtracking. It records its progress by coloring vertices white, gray, or black, as **Depth-First Search** did. Indeed, the same colors and definitions apply. To compare directly with **Depth-First Search**, we can take a snapshot of **Breadth-First Search** executing on the same graph used earlier in Figure 6-6 after it colors its fifth vertex black (vertex 2) as shown in Figure 6-10. At the point shown in the figure, the search has colored black the source vertex *s*, vertices that are one edge away from *s*—{ 1, 6, and 8 }—and vertex 2, which is two edges away from *s*.

The remaining vertices two edges away from *s*—{ 3, 5, 7, 14 }—are all in the queue *Q* waiting to be processed. Some vertices three edges away from *s* have also been visited—{10,11}—and are at the tail of the queue. Note that all vertices within the queue are colored gray, reflecting their active status.

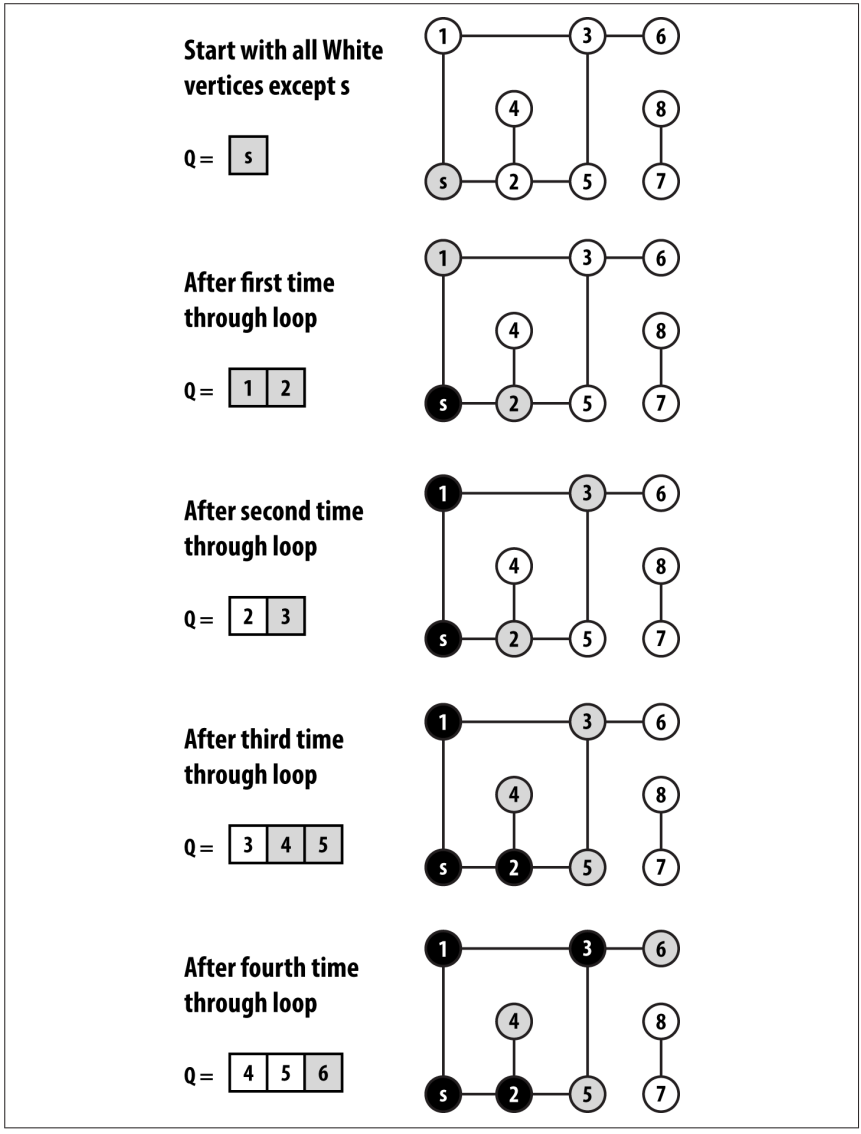
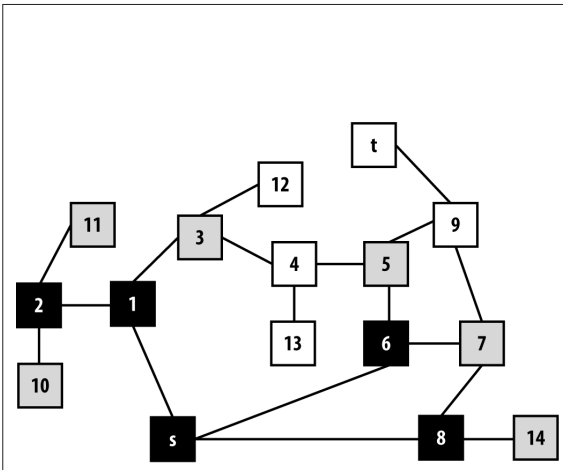


Figure 6-9. Breadth-First Search example

Input/Output

The input is a graph $G = (V, E)$ and a source vertex $s \in V$ representing the start location.

Breadth-First Search produces two computed arrays. `dist[v]` records the number of edges in a shortest path from *s* to *v*. `pred[v]` records the predecessor vertex of *v* based on the breadth-first search ordering. The `pred[]` values record the breadth-first tree search result; if the original graph is unconnected, then all vertices *w* unreachable from *s* have a `pred[w]` value of –1.



v	pred	dist
s	-1	0
1	s	1
2	1	2
3	1	2
4	*	*
5	6	2
6	s	1
7	6	2
8	s	1
9	*	*
10	2	3
11	2	3
12	*	*
13	*	*
14	8	2
t	*	*

Figure 6-10. Breadth-First Search progress on graph after five vertices are colored black

Context

Breadth-First Search stores the vertices being processed in a queue, thus there is O(*V*) storage. **Breadth-First Search** is guaranteed to find a shortest path (there may be ties) in graphs whose vertices are generated "on the fly" (as will be seen in Chapter 7). Indeed, all paths in the generated breadth-first tree are shortest paths from *s* in terms of edge count.

Solution

A sample C++ solution is shown in Example 6-2. **Breadth-First Search** stores its state in a queue, and therefore there are no recursive function calls.

Example 6-2. Breadth-First Search implementation

```
/**
 * Perform breadth-first search on graph from vertex s, and compute BFS
 * distance and pred vertex for all vertices in the graph.
 */
void bfsSearch (Graph const &graph, int s,             /* in */
                vector<int> &dist, vector<int> &pred)  /* out */
{
  // initialize dist and pred to mark vertices as unvisited. Begin at s
  // and mark as Gray since we haven't yet visited its neighbors.
  const int n = graph.numVertices();
  pred.assign(n, -1);
  dist.assign(n, numeric_limits<int>::max());
  vector<vertexColor> color (n, White);

  dist[s] = 0;
  color[s] = Gray;

  queue<int> q;
  q.push(s);
  while (!q.empty()) {
    int u = q.front();

    // Explore neighbors of u to expand the search horizon
    for (VertexList::const_iterator ci = graph.begin (u);
         ci != graph.end (u); ++ci) {
      int v = ci->first;
      if (color[v] == White) {
        dist[v] = dist[u]+1;
        pred[v] = u;
        color[v] = Gray;
        q.push(v);
      }
    }

    q.pop();
    color[u] = Black;
  }
}
```

Analysis

During initialization, **Breadth-First Search** updates information for all vertices, with performance O(V). When a vertex is first visited (and colored gray), it is inserted into the queue, and no vertex is added twice. Since the queue can add and remove elements in constant time, the cost of managing the queue is O(V). Finally, each vertex is dequeued exactly once and its adjacent vertices are traversed exactly once. The sum total of the edge loops, therefore, is bounded by the total number of edges, or O(E). Thus, the total performance is O($V + E$).

Breadth-First Search Summary

Best, Average, Worst: O(V+E)

```
breadthFirstSearch (G, s)
  foreach v in V do
    pred[v] = -1
    dist[v] = ∞
    color[v] = White ❶
  color[s] = Gray
  dist[s] = 0
  Q = empty Queue ❷
  enqueue (Q, s)

  while Q is not empty do
    u = head(Q)
    foreach neighbor v of u do
      if color[v] = White then
        dist[v] = dist[u] + 1
        pred[v] = u
        color[v] = Gray
        enqueue (Q, v)
    dequeue (Q)
    color[u] = Black ❸
end
```

❶ Initially all vertices are marked as not visited.

❷ Queue maintains collection of gray nodes that are visited.

❸ Once all neighbors are visited, this vertex is done.

Single-Source Shortest Path

Suppose you want to fly a private plane on the shortest path from St. Johnsbury, VT, to Waco, TX. Assume you know the distances between the airports for all pairs of cities and towns that are reachable from each other in one nonstop flight of your plane. The best-known algorithm to solve this problem, **Dijkstra's Algorithm**, finds the shortest path from St. Johnsbury to all other airports, although the search may be halted once the shortest path to Waco is known.

In this example, we minimize the distance traversed. In other applications we might replace distance with time (e.g., deliver a packet over a network as quickly as possible) or with cost (e.g., find the cheapest way to fly from St. Johnsbury to Waco). Solutions to these problems also correspond to shortest paths.

Dijkstra's Algorithm relies on a data structure known as a *priority queue* (PQ). A PQ maintains a collection of items, each of which has an associated integer *priority* that represents the importance of an item. A PQ allows one to insert an item, x, with

its associated priority, *p*. Lower values of *p* represent items of higher importance. The fundamental operation of a PQ is *getMin*, which returns the item in PQ whose priority value is lowest (or in other words, is the most important item). Another operation, *decreasePriority*, may be provided by a PQ that allows us to locate a specific item in the PQ and reduce its associated *priority* value (which increases its importance) while leaving the item in the PQ.

Dijkstra's Algorithm Summary

Best, Average, Worst: O((*V*+*E*)log V*)**

```
singleSourceShortest (G, s)
  PQ = empty Priority Queue
  foreach v in V do
    dist[v] = ∞ ❶
    pred[v] = -1

  dist[s] = 0
  foreach v in V do ❷
    insert (v, dist[v]) into PQ

  while PQ is not empty do
    u = getMin(PQ) ❸
    foreach neighbor v of u do
      w = weight of edge (u, v)
      newLen = dist[u] + w
      if newLen < dist[v] then ❹
        decreasePriority (PQ, v, newLen)
        dist[v] = newLen
        pred[v] = u
end
```

❶ Initially all vertices are considered to be unreachable.

❷ Populate PQ with vertices by shortest path distance.

❸ Remove vertex that has shortest distance to source.

❹ If discovered a shorter path from *s* to *v*, record and update PQ.

The source vertex, *s*, is known in advance and `dist[v]` is set to ∞ for all vertices other than *s*, whose `dist[s]` = 0. These vertices are all inserted into the priority queue, `PQ`, with priority equal to `dist[v]`; thus *s* will be the first vertex removed from `PQ`. At each iteration, **Dijkstra's Algorithm** removes a vertex from `PQ` that is closest to *s* of all remaining unvisited vertices in `PQ`. The vertices in `PQ` are potentially updated to reflect a closer distance, given the visited vertices seen so far, as shown in Figure 6-11. After *V* iterations, `dist[v]` contains the shortest distance from *s* to all vertices $v \in V$.

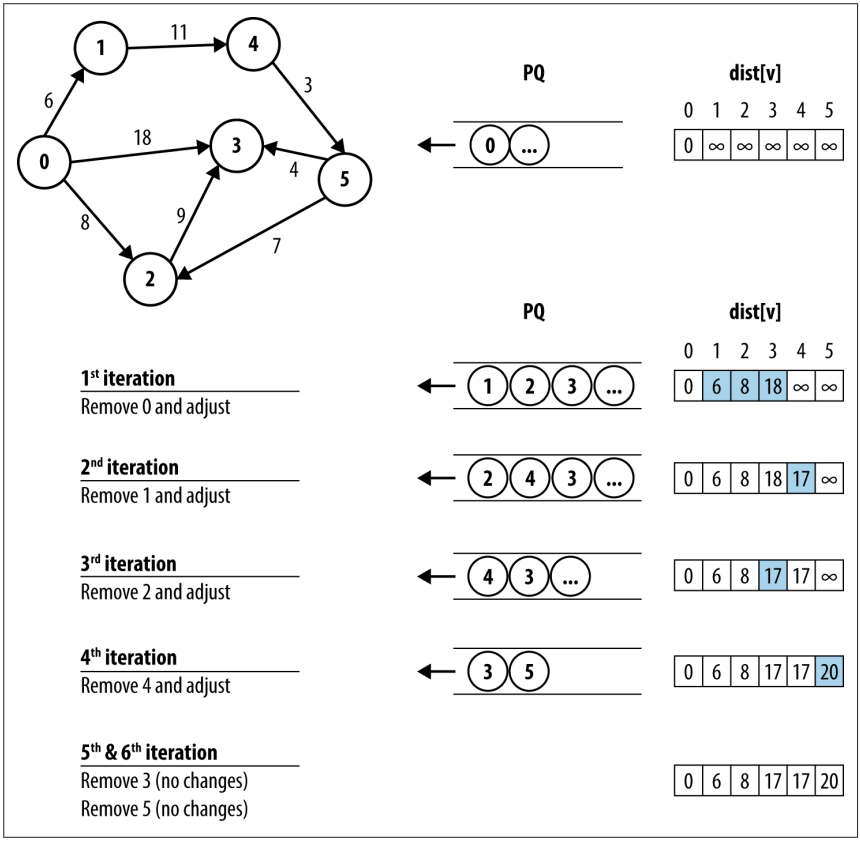
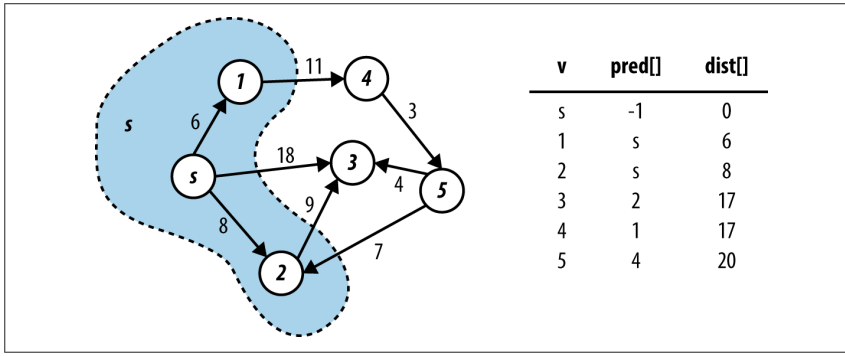


Figure 6-11. Dijkstra's Algorithm example

Dijkstra's Algorithm conceptually operates in a greedy fashion by expanding a set of vertices, *S*, for which the shortest path from a designated source vertex, *s*, to every vertex $v \in S$ is known, *but only using paths that include vertices in S*. Initially, *S* equals the set {*s*}. To expand *S*, as shown in Figure 6-12, **Dijkstra's Algorithm** finds the vertex $v \in V - S$ (i.e., the vertices outside the shaded region) whose distance to *s* is smallest, and follows *v*'s edges to see whether a shorter path exists to some other vertex. After processing v_2, for example, the algorithm determines that the distance from *s* to v_3 containing only vertices in *S* is really 17 through the path $<s, v_2, v_3>$. Once *S* equals *V*, the algorithm completes and the final result is depicted in Figure 6-12.

Input/Output

The input is a directed, weighted graph $G = (V, E)$ and a source vertex $s \in V$. Each edge $e = (u, v)$ has an associated non-negative weight in the graph.



v	pred[]	dist[]
s	-1	0
1	s	6
2	s	8
3	2	17
4	1	17
5	4	20

Figure 6-12. Dijkstra's Algorithm expands the set S

Dijkstra's Algorithm produces two computed arrays. The primary result is the array `dist[]` of values representing the distance from source vertex *s* to each vertex in the graph. The secondary result is the array `pred[]` that can be used to rediscover the actual shortest paths from vertex *s* to each vertex in the graph.

The edge weights are non-negative (i.e., greater than or equal zero); if this assumption is not true, then `dist[]` may contain invalid results. Even worse, **Dijkstra's Algorithm** will loop forever if a cycle exists whose sum of all weights is less than zero.

Solution

As **Dijkstra's Algorithm** executes, `dist[v]` represents the maximum length of the shortest path found from the source *s* to *v* using only vertices visited within the set *S*. Also, for each $v \in S$, `dist[v]` is correct. Fortunately, **Dijkstra's Algorithm** does not need to create and maintain the set *S*. It initially constructs a set containing the vertices in *V*, and then it removes vertices one at a time from the set to compute proper `dist[v]` values; for convenience, we continue to refer to this ever-shrinking set as *V*-*S*. **Dijkstra's Algorithm** terminates when all vertices are either visited or are shown to not be reachable from the source vertex *s*.

In the C++ solution shown in Example 6-3, a *binary heap* stores the vertices in the set *V*-*S* as a priority queue because, in constant time, we can locate the vertex with *smallest priority* (i.e., the vertex's distance from *s*). Additionally, when a shorter path from *s* to *v* is found, `dist[v]` is decreased, requiring the heap to be modified. Fortunately, the *decreasePriority* operation (in a binary heap, it is known as `decreaseKey`) can be performed in O(*log q*) time in the worst case, where *q* is the number of vertices in the binary heap, which will always be less than or equal to the number of vertices, *V*.

Example 6-3. Dijkstra's Algorithm implementation

```
/** Given directed, weighted graph, compute shortest distance to
 * vertices and record predecessor links for all vertices. */
void singleSourceShortest (Graph const &g, int s,  /* in */
                           vector<int> &dist,      /* out */
                           vector<int> &pred) {    /* out */
  // initialize dist[] and pred[] arrays. Start with vertex s by
  // setting dist[] to 0. Priority Queue PQ contains all v in G.
  const int n = g.numVertices();
  pred.assign (n, -1);
  dist.assign (n, numeric_limits<int>::max());
  dist[s] = 0;
  BinaryHeap pq(n);
  for (int u = 0; u < n; u++) { pq.insert (u, dist[u]); }

  // find vertex in ever shrinking set, V-S, whose dist[] is smallest.
  // Recompute potential new paths to update all shortest paths
  while (!pq.isEmpty()) {
    int u = pq.smallest();

    // For neighbors of u, see if newLen (best path from s->u + weight
    // of edge u->v) is better than best path from s->v. If so, update
    // in dist[v] and readjust binary heap accordingly. Compute using
    // longs to avoid overflow error.
    for (VertexList::const_iterator ci = g.begin (u);
         ci != g.end (u); ++ci) {
      int v = ci->first;
      long newLen = dist[u];
      newLen += ci->second;
      if (newLen < dist[v]) {
        pq.decreaseKey (v, newLen);
        dist[v] = newLen;
        pred[v] = u;
      }
    }
  }
}
```

Arithmetic error may occur if the sum of the individual edge weights exceeds `numeric_limits<int>::max()` (although individual values do not). To avoid this situation, compute `newLen` using a long data type.

Analysis

In the implementation of **Dijkstra's Algorithm** in Example 6-3, the `for` loop that constructs the initial priority queue performs the insert operation V times, resulting in performance O(V *log* V). In the remaining `while` loop, each edge is visited once, and thus `decreaseKey` is called no more than E times, which contributes O(E *log* V) time. Thus, the overall performance is O(($V + E$) *log* V).

Dijkstra's Algorithm for Dense Graphs

There is a version of **Dijkstra's Algorithm** suitable for dense graphs represented using an adjacency matrix. The C++ implementation found in Example 6-4 no longer needs a priority queue and it is optimized to use a two dimensional array to contain the adjacency matrix. The efficiency of this version is determined by considering how fast the smallest `dist[]` value in $V - S$ can be retrieved. The `while` loop is executed V times, since S grows one vertex at a time. Finding the smallest `dist[u]` in $V - S$ inspects all V vertices. Note that each edge is inspected exactly once in the inner loop within the `while` loop. Since E can never be larger than V^2, the total running time of this version is O (V^2).

Dijkstra's Algorithm for Dense Graphs Summary

Best, Average, Worst: O($V^2 + E$)

```
singleSourceShortest (G, s)
  foreach v in V do
    dist[v] = ∞ ❶
    pred[v] = -1
    visited[v] = false
  dist[s] = 0

  while some unvisited vertex v has dist[v] < ∞ do ❷
    u = find dist[u] that is smallest of unvisited vertices ❸
    if dist[u] = ∞ then return
    visited[u] = true

    foreach neighbor v of u do
      w = weight of edge (u, v)
      newLen = dist[u] + w
      if newLen < dist[v] then ❹
        dist[v] = newLen
        pred[v] = u
end
```

❶ Initially all vertices are considered to be unreachable.

❷ Stop if all unvisited vertices v have dist[v] = ∞.

❸ Find the vertex that has the shortest distance to source.

❹ If a shorter path is discovered from s to v, record new length.

Because of the adjacency matrix structure, this variation no longer needs a priority queue; instead, at each iteration it selects the unvisited vertex with smallest `dist[]` value. Figure 6-13 demonstrates the execution on a small graph.

Example 6-4. Optimized Dijkstra's Algorithm for dense graphs

```
/** Given int[][] of edge weights as adjacency matrix, compute shortest
 * distance to all vertices in graph (dist) and record predecessor
 * links for all vertices (pred) */
void singleSourceShortestDense (int n, int ** const weight, int s, /* in */
                                int *dist, int *pred) {           /* out */
  // initialize dist[] and pred[] arrays. Start with vertex s by setting
  // dist[] to 0. All vertices are unvisited.
  bool *visited = new bool[n];
  for (int v = 0; v < n; v++) {
    dist[v] = numeric_limits<int>::max();
    pred[v] = -1;
    visited[v] = false;
  }
  dist[s] = 0;

  // find shortest distance from s to unvisited vertices.  Recompute
  // potential new paths to update all shortest paths.
  while (true) {
    int u = -1;
    int sd = numeric_limits<int>::max();
    for (int i = 0; i < n; i++) {
      if (!visited[i] && dist[i] < sd) {
        sd = dist[i];
        u = i;
      }
    }
    if (u == -1) { break; } // exit if no new paths found

    // For neighbors of u, see if best path-length from s->u + weight of
    // edge u->v is better than best path from s->v. Compute using longs.
    visited[u] = true;
    for (int v = 0; v < n; v++) {
      int w = weight[u][v];
      if (v == u) continue;

      long newLen = dist[u];
      newLen += w;
      if (newLen < dist[v]) {
        dist[v] = newLen;
        pred[v] = u;
      }
    }
  }
  delete [] visited;
}
```

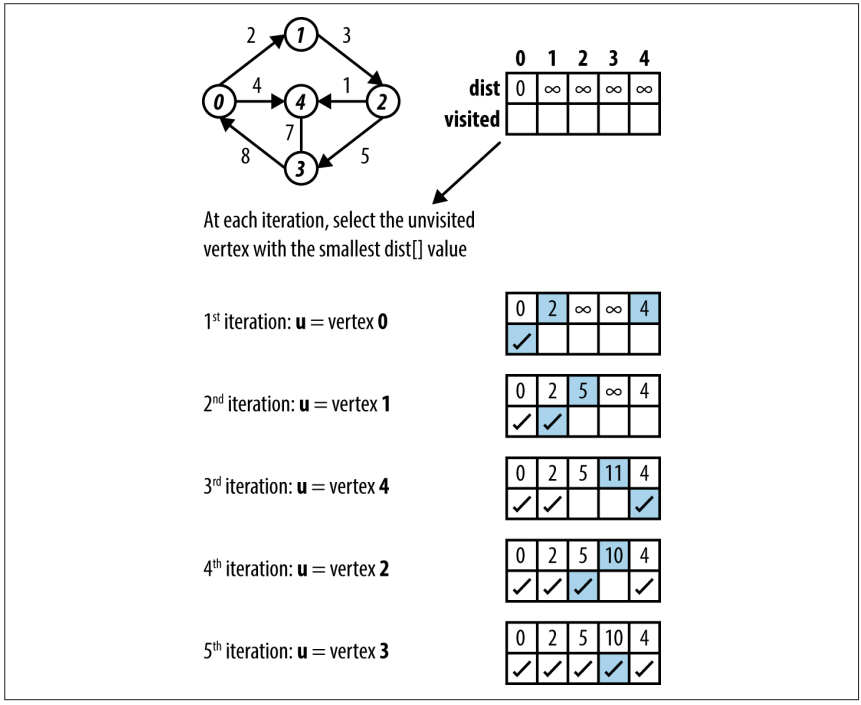


Figure 6-13. Dijkstra's Algorithm dense graph example

Variations

We may seek the most reliable path to send a message from one point to another through a network where we know the probability that any leg of a transmission delivers the message correctly. The probability of any path (i.e., a sequence of legs) delivering a message correctly is the product of all the probabilities along the path. Using the same technique that makes multiplication possible on a slide rule, we can replace the probability on each edge with the negative value of the logarithm of the probability. The shortest path in this new graph corresponds to the most reliable path in the original graph.

Dijkstra's Algorithm cannot be used when edge weights are negative. However, **Bellman–Ford** can be used as long as there is no cycle whose edge weights sum to a value less than zero. The concept of "shortest path" is meaningless when such a cycle exists. Although the sample graph in Figure 6-14 contains a cycle involving vertices {1,3,2}, the edge weights are positive, so **Bellman–Ford** will work.

A C++ implementation of Bellman-Ford is shown in Example 6-5.

Bellman–Ford Summary

Best, Average, Worst: O(*V*E*)

```
singleSourceShortest (G, s)
  foreach v in V do
    dist[v] = ∞ ❶
    pred[v] = -1
  dist[s] = 0

  for i = 1 to n do
    foreach edge (u,v) in E do
      newLen = dist[u] + weight of edge (u,v)
      if newLen < dist[v] then ❷
        if i = n then report "Negative Cycle"
        dist[v] = newLen
        pred[v] = u
end
```

❶ Initially all vertices are considered to be unreachable.

❷ If a shorter path is discovered from *s* to *v*, record new length.

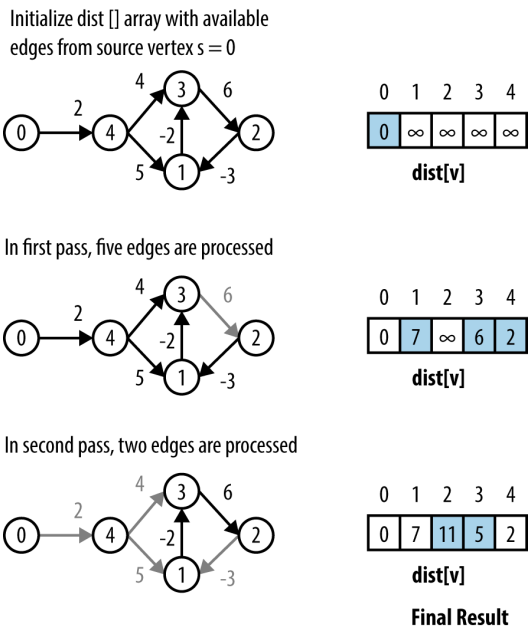


Figure 6-14. Bellman–Ford example

Example 6-5. Bellman–Ford algorithm for single-source shortest path

```
/**
 * Given directed, weighted graph, compute shortest distance to vertices
 * in graph (dist) and record predecessor links for all vertices (pred) to
 * be able to re-create these paths. Graph weights can be negative so long
 * as there are no negative cycles.
 */
void singleSourceShortest (Graph const &graph, int s,              /* in */
                           vector<int> &dist, vector<int> &pred) { /* out */
  // initialize dist[] and pred[] arrays.
  const int n = graph.numVertices();
  pred.assign (n, -1);
  dist.assign (n, numeric_limits<int>::max());
  dist[s] = 0;

  // After n-1 times we can be guaranteed distances from s to all
  // vertices are properly computed to be shortest. So on the nth
  // pass, a change to any value guarantees there is a negative cycle.
  // Leave early if no changes are made.
  for (int i = 1; i <= n; i++) {
    bool failOnUpdate = (i == n);
    bool leaveEarly = true;

    // Process each vertex, u, and its respective edges to see if
    // some edge (u,v) realizes a shorter distance from s->v by going
    // through s->u->v. Use longs to prevent overflow.
    for (int u = 0; u < n; u++) {
      for (VertexList::const_iterator ci = graph.begin (u);
           ci != graph.end (u); ++ci) {
        int v = ci->first;
        long newLen = dist[u];
        newLen += ci->second;
        if (newLen < dist[v]) {
          if (failOnUpdate) { throw "Graph has negative cycle"; }
          dist[v] = newLen;
          pred[v] = u;
          leaveEarly = false;
        }
      }
    }
    if (leaveEarly) { break; }
  }
}
```

Intuitively **Bellman–Ford** operates by making n sweeps over a graph that check to see if any edge (u, v) is able to improve on the computation for `dist[v]` given `dist[u]` and the weight of the edge over (u, v). At least $n - 1$ sweeps are needed, for example, in the extreme case that the shortest path from s to some vertex v goes through all vertices in the graph. Another reason to use $n - 1$ sweeps is that the

edges can be visited in an arbitrary order, and this ensures all reduced paths have been found.

Bellman–Ford is thwarted only when there exists a negative cycle of directed edges whose total sum is less than zero. To detect such a negative cycle, we execute the primary processing loop *n* times (one more than necessary), and if there is an adjustment to some `dist[]` value, a negative cycle exists. The performance of **Bellman–Ford** is $O(V*E)$, as clearly seen by the nested `for` loops.

Comparing Single-Source Shortest-Path Options

The following summarizes the expected performance of the three algorithms by computing a rough cost estimate:

- **Bellman–Ford**: $O(V*E)$
- **Dijkstra's Algorithm** for dense graphs: $O(V^2 + E)$
- **Dijkstra's Algorithm** with priority queue: $O((V + E)*log\ V)$

We compare these algorithms under different scenarios. Naturally, to select the one that best fits your data, you should benchmark the implementations as we have done. In the following tables, we execute the algorithms 10 times and discard the best- and worst-performing runs; the tables show the average of the remaining eight runs.

Benchmark Data

It is difficult to generate random graphs. In Table 6-1, we show the performance on generated graphs with $|V| = k^2 + 2$ vertices and $|E| = k^3–k^2 + 2k$ edges in a highly stylized graph construction (for details, see the code implementation in the repository). Note that the number of edges is roughly $n^{1.5}$ where n is the number of vertices in V. The best performance comes from using the priority queue implementation of **Dijsktra's Algorithm** but **Bellman–Ford** is not far behind. Note how the variations optimized for dense graphs perform poorly.

Table 6-1. Time (in seconds) to compute single-source shortest path on benchmark graphs

V	E	Dijkstra's Algorithm with PQ	Optimized Dijkstra's Algorithm for DG	Bellman–Ford
6	8	0.000002	0.000002	0.000001
18	56	0.000004	0.000003	0.000001
66	464	0.000012	0.000018	0.000005
258	3,872	0.00006	0.000195	0.000041
1,026	31,808	0.000338	0.0030	0.000287

V	E	Dijkstra's Algorithm with PQ	Optimized Dijkstra's Algorithm for DG	Bellman–Ford
4,098	258,176	0.0043	0.0484	0.0076
16,386	2,081,024	0.0300	0.7738	0.0535

Dense Graphs

For dense graphs, E is on the order of $O(V^2)$; for example, in a complete graph of $n = |V|$ vertices that contains an edge for every pair of vertices, there are $n^*(n - 1)/2$ edges. Using **Bellman–Ford** on such dense graphs is not recommended, since its performance degenerates to $O(V^3)$. The set of dense graphs reported in Table 6-2 is taken from a set of publicly available data sets used by researchers investigating the Traveling Salesman Problem (TSP). We executed 100 trials and discarded the best and worst performances; the table contains the average of the remaining 98 trials. Although there is little difference between the priority queue and dense versions of **Dijsktra's Algorithm**, there is a vast improvement in the optimized **Dijsktra's Algorithm**, as shown in the table. In the final column we show the performance time for **Bellman–Ford** for the same problems, but these results are the averages of only five executions because the performance degrades so sharply. The lesson to draw from the last column is that the absolute performance of **Bellman–Ford** on sparse graphs seems to be quite reasonable, but when compared relatively to its peers on dense graphs, we see clearly that it is the wrong algorithm to use (unless there are edges with negative weights, in which case this algorithm must be used).

Table 6-2. Time (in seconds) to compute single-source shortest path on dense graphs

V	E	Dijkstra's Algorithm with PQ	Optimized Dijkstra's Algorithm for DG	Bellman–Ford
980	479,710	0.0681	0.0050	0.1730
1,621	1,313,010	0.2087	0.0146	0.5090
6,117	18,705,786	3.9399	0.2056	39.6780
7,663	29,356,953	7.6723	0.3295	40.5585
9,847	48,476,781	13.1831	0.5381	78.4154
9,882	48,822,021	13.3724	0.5413	42.1146

Sparse graphs

Large graphs are frequently sparse, and the results in Table 6-3 confirm that one should use the **Dijsktra's Algorithm** with a priority queue rather than the implementation crafted for dense graphs; note how the implementation for dense graphs is noticeably slower. The rows in the table are sorted by the number of edges in the sparse graphs, since that appears to be the determining cost factor in the results.

Table 6-3. Time (in seconds) to compute single-source shortest path on large sparse graphs

V	E	Density	Dijkstra's Algorithm with PQ	Optimized Dijkstra's Algorithm for DG
3,403	137,845	2.4%	0.0102	0.0333
3,243	294,276	5.6%	0.0226	0.0305
19,780	674,195	0.34%	0.0515	1.1329

All-Pairs Shortest Path

Instead of finding the shortest path from a single source, we often seek the shortest path between any two vertices (v_i, v_j); there may be several paths with the same total distance. The fastest solution to this problem uses the Dynamic Programming technique introduced in Chapter 3.

There are two interesting features of dynamic programming:

- It stores the solution to small, constrained versions of the problem.
- Although we seek an optimal answer to a problem, it is easier to compute the *value* of an optimal answer rather than the answer itself. In our case, we compute, for each pair of vertices (v_i, v_j), the length of a shortest path from v_i to v_j and perform additional computation to recover the actual path. In the following pseudocode, k, u, and v each represent a potential vertex of G.

Floyd–Warshall computes an n-by-n matrix `dist` such that for all pairs of vertices (v_i, v_j), `dist[i][j]` contains the length of a shortest path from v_i to v_j.

Figure 6-15 demonstrates an example of **Floyd-Warshall** on the sample graph from Figure 6-13. As you can confirm, the first row of the computed matrix is the same as the computed vector from Figure 6-13. **Floyd-Warshall** computes the shortest path between all pairs of vertices.

Input/Output

The input is a directed, weighted graph $G = (V, E)$. Each edge $e = (u, v)$ has an associated positive (i.e., greater than zero) weight in the graph.

Floyd–Warshall computes a matrix `dist[][]` representing the shortest distance from each vertex u to every vertex in the graph (including itself). Note that if `dist[u][v]` is ∞, then there is no path from u to v. The actual shortest path between any two vertices can be computed from a second matrix, `pred[][]`, also computed by the algorithm.

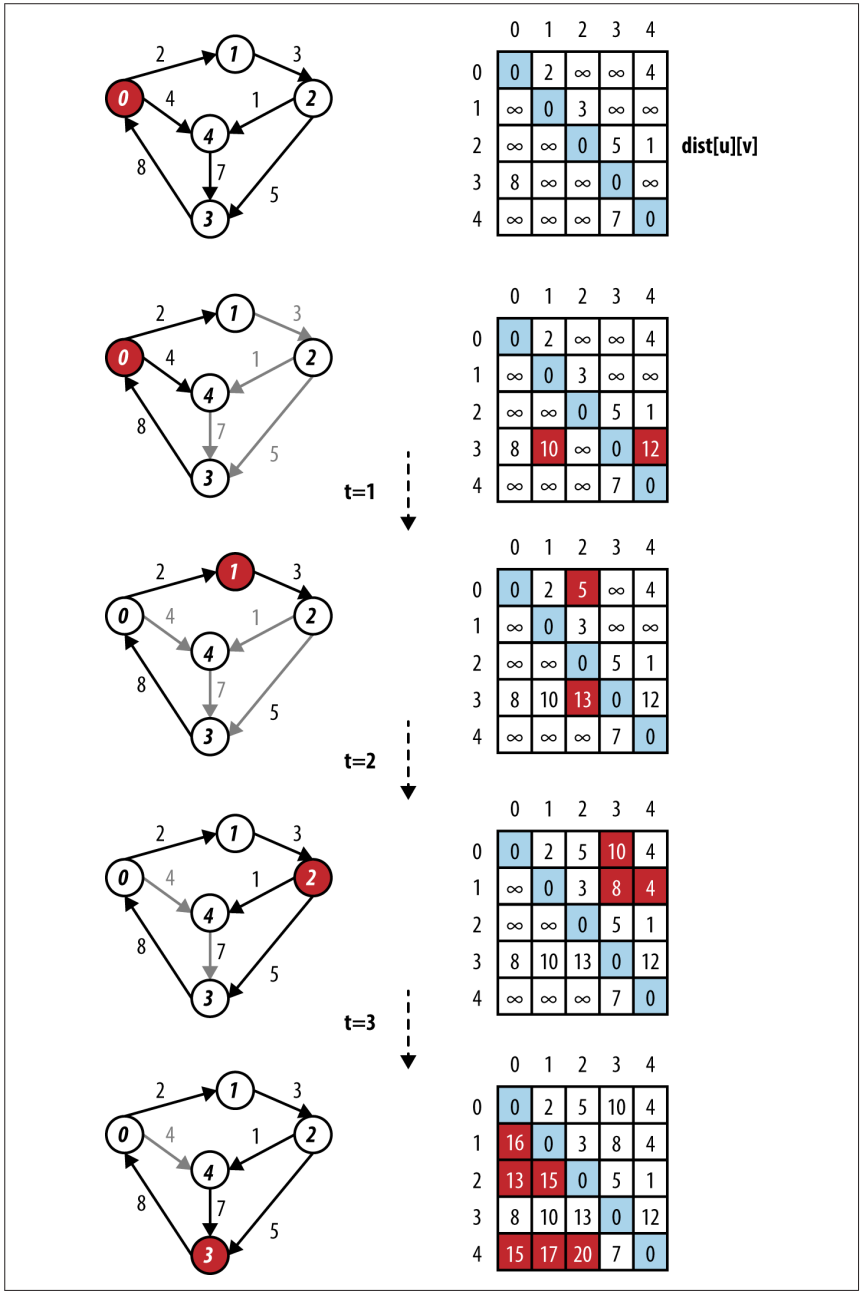


Figure 6-15. Floyd–Warshall example

Floyd–Warshall Summary

Best, Average, Worst: O(V^3)

```
allPairsShortestPath (G)
  foreach u in V do
    foreach v in V do
      dist[u][v] = ∞ ❶
      pred[u][v] = -1
    dist[u][u] = 0
    foreach neighbor v of u do
      dist[u][v] = weight of edge (u,v)
      pred[u][v] = u

  foreach k in V do
    foreach u in V do
      foreach v in V do
        newLen = dist[u][k] + dist[k][v]
        if newLen < dist[u][v] then ❷
          dist[u][v] = newLen
          pred[u][v] = pred[k][v] ❸
end
```

❶ Initially all vertices are considered to be unreachable.

❷ If a shorter path is discovered from *s* to *v* record new length.

❸ Record the new predecessor link.

Solution

A Dynamic Programming approach computes, in order, the results of simpler subproblems. Consider the edges of *G*: these represent the length of the shortest path between any two vertices *u* and *v* that **does not include any other vertex**. Thus, the `dist[u][v]` matrix is set initially to ∞ and `dist[u][u]` is set to zero (to confirm that there is no cost for the path from a vertex *u* to itself). Finally, `dist[u][v]` is set to the weight of every edge $(u, v) \in E$. At this point, the `dist` matrix contains the best computed shortest path (so far) for each pair of vertices, (u, v).

Now consider the next larger subproblem, namely, computing the length of the shortest path between any two vertices *u* and *v* that **might also include** v_1. Dynamic Programming will check each pair of vertices (u, v) to check whether the path $\{u, v_1, v\}$ has a total distance smaller than the best score. In some cases, `dist[u][v]` is still ∞ because there was no information on any path between *u* and *v*. At other times, the sum of the paths from *u* to v_1 and then from v_1 to *v* is better than the current distance, and the algorithm records that total in `dist[u][v]`. The next larger subproblem tries to compute the length of the shortest path between any two vertices *u* and *v* that **might also include** v_1 or v_2. Eventually the algorithm increases

these subproblems until the result is a `dist[u][v]` matrix that contains the shortest path between any two vertices *u* and *v* that might also include any vertex in the graph.

Floyd–Warshall computes the key optimization check whether `dist[u][k] + dist[k][v] < dist[u][v]`. Note that the algorithm also computes a `pred[u][v]` matrix that "remembers" that the newly computed shorted path from *u* to *v* must go through vertex *k*. The surprisingly brief solution is shown in Example 6-6.

Example 6-6. Floyd–Warshall algorithm for computing all-pairs shortest path

```
void allPairsShortest (Graph const &graph,     /* in */
        vector< vector<int> > &dist,           /* out */
        vector< vector<int> > &pred) {         /* out */
  int n = graph.numVertices();

  // Initialize dist[][] with 0 on diagonals, INFINITY where no edge
  // exists, and the weight of edge (u,v) placed in dist[u][v]. pred
  // initialized in corresponding way.
  for (int u = 0; u < n; u++) {
    dist[u].assign (n, numeric_limits<int>::max());
    pred[u].assign (n, -1);
    dist[u][u] = 0;
    for (VertexList::const_iterator ci = graph.begin (u);
         ci != graph.end (u); ++ci) {
      int v = ci->first;
      dist[u][v] = ci->second;
      pred[u][v] = u;
    }
  }

  for (int k = 0; k < n; k++) {
    for (int i = 0; i < n; i++) {
      if (dist[i][k] == numeric_limits<int>::max()) { continue; }

      // If an edge is found to reduce distance, update dist[][].
      // Compute using longs to avoid overflow of Infinity distance.
      for (int j = 0; j < n; j++) {
        long newLen = dist[i][k];
        newLen += dist[k][j];

        if (newLen < dist[i][j]) {
          dist[i][j] = newLen;
          pred[i][j] = pred[k][j];
        }
      }
    }
  }
}
```

The function shown in Example 6-7 constructs an actual shortest path (there may be more than one) from a given *s* to *t*. It works by recovering predecessor information from the `pred` matrix.

Example 6-7. Code to recover shortest path from computed pred[][]

```
/** Output path as list of vertices from s to t given the pred results
 * from an allPairsShortest execution. Note that s and t must be valid
 * integer vertex identifiers. If no path is found between s and t, then
 * an empty path is returned.  */
void constructShortestPath (int s, int t,    /* in */
        vector< vector<int> > const &pred,   /* in */
        list<int> &path) {                   /* out */
  path.clear();
  if (t < 0 || t >= (int) pred.size() || s < 0 || s >= (int) pred.size()) {
    return;
  }

  // construct path until we hit source 's' or -1 if there is no path.
  path.push_front (t);
  while (t != s) {
    t = pred[s][t];
    if (t == -1) { path.clear (); return; }

    path.push_front (t);
  }
}
```

Analysis

The time taken by **Floyd–Warshall** is dictated by the number of times the minimization function is computed, which is $O(V^3)$, as can be seen from the three nested `for` loops. The `constructShortestPath` function in Example 6-7 executes in $O(E)$ since the shortest path might include every edge in the graph.

Minimum Spanning Tree Algorithms

Given an undirected, connected graph $G = (V, E)$, we might be concerned with finding a subset ST of edges from E that "span" the graph because it connects all vertices. If we further require that the total weight of the edges in ST is the minimal across all possible spanning trees, then we are interested in finding a minimum spanning tree (MST).

Prim's Algorithm shows how to construct an MST from such a graph by using a Greedy approach in which each step of the algorithm makes forward progress toward a solution without reversing earlier decisions. **Prim's Algorithm** grows a spanning tree T one edge at a time until an MST results (and the resulting spanning tree is provably minimum). It randomly selects a start vertex $s \in v$ to belong to a growing set S and ensures that T forms a tree of edges in S. **Prim's Algorithm** is

greedy in that it incrementally adds edges to T until an MST is computed. The intuition behind the algorithm is that the edge (u, v) with lowest weight between $u \in S$ and $v \in V\text{-}S$ must belong to the MST. When such an edge (u, v) with lowest weight is found, it is added to T and the vertex v is added to S.

The algorithm uses a priority queue to store the vertices $v \in V - S$ with an associated *priority key* equal to the lowest weight of some edge (u, v) where $u \in S$. This key value reflects the priority of the element within the priority queue; smaller values are of higher importance.

Prim's Algorithm Summary

Best, Average, Worst: O(*(V + E)*log V*)

```
computeMST (G)
  foreach v in V do
    key[v] = ∞ ❶
    pred[v] = -1
  key[0] = 0
  PQ = empty Priority Queue
  foreach v in V do
    insert (v, key[v]) into PQ

  while PQ is not empty do
    u = getMin(PQ) ❷
    foreach edge(u,v) in E do
      if PQ contains v then
        w = weight of edge (u,v)
        if w < key[v] then ❸
          pred[v] = u
          key[v] = w
          decreasePriority (PQ, v, w)
end
```

❶ Initially all vertices are considered to be unreachable.

❷ Find vertex in V with lowest computed distance.

❸ Revise cost estimates for v and record MST edge in $pred[v]$.

Figure 6-16 demonstrates the behavior of **Prim's Algorithm** on a small undirected graph. The priority queue is ordered based on the distance from the vertices in the queue to any vertex already contained in the MST.

Input/Output

The input is an undirected graph $G = (V, E)$.

The output is an MST encoded in the `pred[]` array. The *root* of the MST is the vertex whose `pred[v]` = –1.

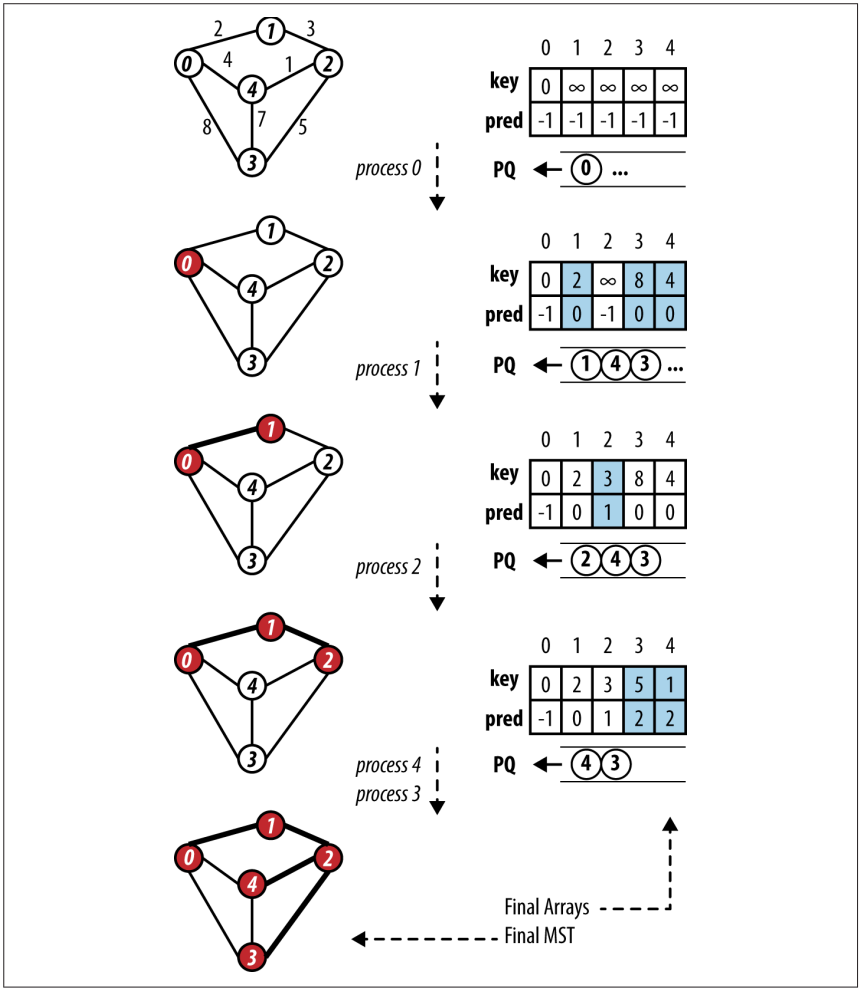


Figure 6-16. Prim's Algorithm example

Solution

The C++ solution shown in Example 6-8 relies on a binary heap to provide the implementation of the priority queue that is central to **Prim's Algorithm**. Ordinarily, using a binary heap would be inefficient because of the check in the main loop for whether a particular vertex is a member of the priority queue (an operation not supported by binary heaps). However, the algorithm ensures each vertex is removed from the priority queue only after being processed by the program, thus it maintains a status array `inQueue[]` that is updated whenever a vertex is extracted from the priority queue. In another implementation optimization, it maintains an external array `key[]` recording the current priority key for each vertex in the queue, which again eliminates the need to search the priority queue for a given vertex.

Prim's Algorithm randomly selects one of the vertices to be the starting vertex, *s*. When the minimum vertex, *u*, is removed from the priority queue PQ and "added" to the visited set *S*, the algorithm uses existing edges between *S* and the growing spanning tree *T* to reorder the elements in PQ. Recall that the *decreasePriority* operation moves an element closer to the front of PQ.

Example 6-8. Prim's Algorithm implementation with binary heap

```
/** Given undirected graph, compute MST starting from a randomly
 * selected vertex. Encoding of MST is done using 'pred' entries. */
void mst_prim (Graph const &graph, vector<int> &pred) {
  // initialize pred[] and key[] arrays. Start with arbitrary
  // vertex s=0. Priority Queue PQ contains all v in G.
  const int n = graph.numVertices();
  pred.assign (n, -1);
  vector<int> key(n, numeric_limits<int>::max());
  key[0] = 0;
  BinaryHeap pq(n);
  vector<bool> inQueue(n, true);
  for (int v = 0; v < n; v++) {
    pq.insert (v, key[v]);
  }

  while (!pq.isEmpty()) {
    int u = pq.smallest();
    inQueue[u] = false;

    // Process all neighbors of u to find if any edge beats best distance
    for (VertexList::const_iterator ci = graph.begin (u);
         ci != graph.end (u); ++ci) {
      int v = ci->first;
      if (inQueue[v]) {
        int w = ci->second;
        if (w < key[v]) {
          pred[v] = u;
          key[v] = w;
          pq.decreaseKey (v, w);
        }
      }
    }
  }
}
```

Analysis

The initialization phase of **Prim's Algorithm** inserts each vertex into the priority queue (implemented by a binary heap) for a total cost of O(*V log V*). The `decrease Key` operation in **Prim's Algorithm** requires O(*log q*) performance, where *q* is the number of elements in the queue, which will always be less than *V*. It can be called at most 2**E* times since each vertex is removed once from the priority queue and

each undirected edge in the graph is visited exactly twice. Thus, the total performance is O((*V* + 2**E*)**log V*) or O((*V* + *E*)**log V*).

Variations

Kruskal's Algorithm is an alternative to **Prim's Algorithm**. It uses a "disjoint-set" data structure to build up the minimum spanning tree by processing all edges in the graph in order of weight, starting with the edge with the smallest weight and ending with the edge with the largest weight. **Kruskal's Algorithm** can be implemented in O(*E log V*). Details on this algorithm can be found in (Cormen et al., 2009).

Final Thoughts on Graphs

In this chapter, we have seen that the algorithms behave differently based on whether a graph is sparse or dense. We will now explore this concept further to analyze the break-even point between sparse and dense graphs and understand the impact on storage requirements.

Storage Issues

When using a two-dimensional adjacency matrix to represent potential relationships among *n* elements in a set, the matrix requires n^2 elements of storage, yet there are times when the number of relationships is much smaller. In these cases—known as *sparse* graphs—it may be impossible to store large graphs with more than several thousand vertices because of the limitations of computer memory. Additionally, traversing through large matrices to locate the few edges in sparse graphs becomes costly, and this storage representation prevents efficient algorithms from achieving their true potential.

The adjacency representations discussed in this chapter contain the same information. Suppose, however, you were writing a program to compute the cheapest flights between any pair of cities in the world that are served by commercial flights. The weight of an edge would correspond to the cost of the cheapest direct flight between that pair of cities (assuming airlines do not provide incentives by bundling flights). In 2012, Airports Council International (ACI) reported a total of 1,598 airports worldwide in 159 countries, resulting in a two-dimensional matrix with 2,553,604 entries. The question "how many of these entries has a value?" is dependent on the number of direct flights. ACI reported 79 million "aircraft movements" in 2012, roughly translating to a daily average of 215,887 flights. Even if all of these flights represented an actual direct flight between two unique airports (clearly the number of direct flights will be much smaller), this means the matrix is 92% empty—a good example of a sparse matrix!

Graph Analysis

When applying the algorithms in this chapter, the essential factor that determines whether to use an adjacency list or adjacency matrix is whether the graph is sparse. We compute the performance of each algorithm in terms of the number of vertices

in the graph, V, and the number of edges in the graph, E. As is common in the literature on algorithms, we simplify the presentation of the formulas that represent best, average, and worst case by using V and E within the big-O notation. Thus, $O(V)$ means a computation requires a number of steps that is directly proportional to the number of vertices in the graph. However, the density of the edges in the graph will also be relevant. Thus, $O(E)$ for a sparse graph is on the order of $O(V)$, whereas for a dense graph it is closer to $O(V^2)$.

As we will see, the performance of some algorithms depends on the structure of the graph; one variation might execute in $O((V + E)*log\ V)$ time, while another executes in $O(V^2 + E)$ time. Which one is more efficient? Table 6-4 shows that the answer depends on whether the graph G is sparse or dense. For sparse graphs, $O((V + E)*log\ V)$ is more efficient, whereas for dense graphs $O(V^2 + E)$ is more efficient. The table entry labeled "Break-even graph" identifies the type of graphs for which the expected performance is the same $O(V^2)$ for both sparse and dense graphs; in these graphs, the number of edges is on the order of $O(V^2/log\ v)$.

Table 6-4. Performance comparison of two algorithm variations

Graph type	O((V + E)*logV)	Comparison	$O(V^2 + E)$
Sparse graph: $\|E\|$ is $O(V)$	$O(V\ log\ V)$	is smaller than	$O(V^2)$
Break-even graph: $\|E\|$ is $O(V^2/log\ V)$	$O(V^2 + V\ log\ v) = O(V^2)$	is equivalent to	$O(V^2 + V^2/log\ V) = O(V^2)$
Dense graph: $\|E\|$ is $O(V^2)$	$O(V^2\ log\ V)$	is larger than	$O(V^2)$

References

Cormen, T. H., C. Leiserson, R. Rivest, and C. Stein, *Introduction to Algorithms*. Third Edition. MIT Press, 2009.